

Python Code

```
1  import os
2  import nltk
3  from nltk.corpus import stopwords
4  from nltk.tokenize import word_tokenize
5  from nltk.stem import PorterStemmer
6  from collections import defaultdict
7  import json
8
9  # Download necessary NLTK data
10 nltk.download('punkt')
11 nltk.download('stopwords')
12
13 def preprocess(text):
14     text = text.lower()
15     tokens = word_tokenize(text)
16
17     # Initialize components
18     stop_words = set(stopwords.words('english'))
19     stemmer = PorterStemmer()
20
21     # Process words: stem, filter alphanumeric, and remove stop words
22     words = [
23         stemmer.stem(word)
24         for word in tokens
25         if word.isalnum() and word not in stop_words
26     ]
27     return words
28
29 # Initialize document storage
30 documents = {}
31
32 # Load and preprocess documents from the current directory
33 for filename in os.listdir():
34     if filename.endswith(".txt"):
35         with open(filename, 'r', encoding='utf-8', errors='ignore') as f:
36             text = f.read()
37             documents[filename] = preprocess(text)
38
39 print(f"Total documents loaded: {len(documents)}")
40
41 # Build the Inverted Index
42 inverted_index = defaultdict(set)
43 for doc_id, words in documents.items():
44     # Use set(words) to ensure each term is indexed only once per document
45     for word in set(words):
46         inverted_index[word].add(doc_id)
47
48 # Print index statistics
49 vocab_size = len(inverted_index)
50 print(f"\nVocabulary Size: {vocab_size} words")
51
52 # Print sample index terms
```

```
53 print("\nSample inverted index terms:")
54 for term in list(inverted_index.keys())[:10]:
55     # Use sorted() for clean output of document IDs
56     print(f"{term}: {sorted(list(inverted_index[term]))}")
57
58
59
60
61
62
```